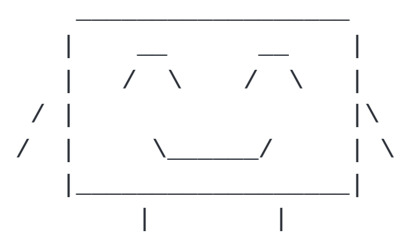
Ooonana OS



Ooonana OS 0.10.0 field guide
Evidence refreshed 2026-11-04; core and installed WSL 0.10.0 candidate; stable unpromoted.

What it is

Ooonana OS is a scratch-built Linux project with its own rootfs, boot flow, installer experiments, WSL export, and custom ooonana package manager.
Debian or Ubuntu are host build tools only. Alpine APKs are imported into Ooonana .pkg repos; the target OS installs Ooonana packages, not live Alpine APKs.
Core 0.10.0 is a supported-base candidate, not a stable release. Imported payloads use one Alpine v3.24 branch. Stable 0.9.9 installations and public update channel remain separate during validation.
Private native VM checksum installation/reinstallation, BIOS/UEFI boot, UUID EFI/swap mounts, nonroot login/sudo and clean shutdown pass. Signed fixture health rollback, prepared-checkpoint power-loss recovery and bounded ext4 full-storage rollback also pass. Physical USB, broader system/kernel upgrades and final ISO acceptance remain separate gates.

Editions

minimal: BusyBox-style rootfs, kernel, GRUB disk, installer ISO, WSL rootfs, command line AI.
full-i3: minimal plus i3, Xorg, searchable application launcher, movable GTK apps, wallpaper, GUI installer, AI desktop app, and full WSL rootfs.

Package install

```
ooonana update
ooonana search nano
ooonana show nano
ooonana get nano --dry-run
ooonana get nano
ooonana files nano
ooonana verify nano
ooonana upgrade
ooonana remove nano
ooonana purge nano
ooonana fix nano --reinstall
```

Cloud repo

Default cloud path is the GitLab Pages direct repository:
https://ooonana.gitlab.io/ooonana-repo
Run ooonana update && ooonana upgrade. Metadata syncs first; package archives download only when install or upgrade needs them.
GitHub Releases remains a backup tarball repository path.

Installer

Full-i3 live ISO boots i3. Installer requires explicit target and dry-run preview. Erase-disk mode offers optional swap; custom root, home, swap and EFI must share target disk.
Installer validates identity before disk writes, locks root, writes filesystem UUIDs and checks retained-file content even when size/timestamp match. These repairs passed actual private installation tests; they do not certify physical USB fault behavior.

USB live modes

Normal live mode mounts ISO and rootfs read-only. It uses a cleared temporary overlay on OOONANA_PERSIST when available on the same boot USB, with RAM fallback.
Persistent live mode uses an ext4 OOONANA_PERSIST partition or a configured ext4 persistence file on the same boot USB. Setup previews and requires confirmation; GRUB selection never silently formats storage. Full writable overlay saves files, settings, radio pairings, packages and system changes.
GRUB boot UUID and same-parent checks protect unrelated media. Cloned UUIDs, unsafe paths, missing/unwritable persistence, and mismatched ISO base identity stop in recovery; no silent RAM fallback or automatic repair.
Other disks may be probed read-only. Live mode writes only to matching boot USB persistence. Only the confirmed installer target can be partitioned or formatted.

First boot

ooonana setup --first-boot --gui can create user, set password, write network config, choose theme, set zram and disk-swap policy, add cloud repo, and mark setup complete.

WSL

For a new test distro only: scripts/install-wsl-distro.sh --distro OoonanaCandidate --tarball /var/tmp/ooonana-os/release/ooonana-full-i3-wsl-rootfs.tar.gz. Never force-reimport over an existing distro as a migration shortcut.
Launch installed desktop:
wsl -d Ooonana -u ooonana -- /usr/local/bin/start-ooonana-i3 --nested
WSL GUI needs WSLg and Xephyr, or an X server with DISPLAY set. Nested mode keeps i3 panel and dock together in one window.
WSL-local adapter preserves the packaged launcher, waits up to 60 seconds for cold Xephyr startup, and selects X11/XCB. Installed WSL uses verified local signed generations under /usr/lib/ooonana/repo; no removable host drive mount is required.
Ubuntu remains the build host; no distro disk is moved.
No AI model weights were downloaded into fresh WSL. OpenVINO app is installed; inference runtime/models need explicit setup. Windows model folders remain untouched.
Readiness checks do not certify interactive input, lock/logout or suspend/resume.

WSL (continued)

Same-base maintenance: ooonana update && ooonana upgrade. Cross-base migration requires verified backup and whole-world upgrade --allow-major from an independent host against an offline target root. Live libc replacement is refused; old saved USB overlays need separate explicit data migration.

AI

Ooonana AI runs as ooonana ai ... or ooonana-ai. It supports cloud providers, offline Intel OpenVINO, ask/chat, tools, tasks, audit, history, status, and a native GUI.
The full-i3 desktop includes a native AI workspace, Settings, Wi-Fi, Bluetooth, Packages and searchable application launcher. OpenVINO overhaul and new runtime/model validation remain deferred.
Pinned Linux runtime, real Korean/English tokenizer, strict RAM refusal, native CPU streaming/cancel and loaded-engine restart passed with small QA models.
Heavyweight/GPU model inference remains pending. Fresh WSL app installation does not imply inference runtime or models are configured.
Load and generation allocation failures return insufficient_memory (HTTP503, or typed SSE error after streaming headers). Reduce context/output length or choose a smaller model; swap is not extra physical RAM.

Desktop and hardware

The desktop uses opaque Graphite/Paper surfaces, amber accents and restrained rounding. Top panel carries system/service status and left-side music. Flat centered dock groups all application windows, with pins, running indicators, exact restore and right-click actions.
Core 0.10.0 retains matching native icons, focus-neutral dock previews, responsive panel spacing, AI loading/RAM/storage indicators, and compact third-party controls that exclude hidden tabs and fullscreen-covered windows. Alt+F4 closes; Alt+F10 toggles fullscreen.
Eighteen exact-package normal/scaled desktop cases pass. Geany file-picker focus, cancel and file opening passed privately at 1280x800 and 1024x768; initial dialogs are bounded and centered. Physical/mixed-DPI/session behavior is not certified by Xvfb.
Ctrl+Shift+Esc opens native Task Manager with processes, performance, and available temperature and fan sensors. Unavailable hardware counters are labeled, not guessed.
Live USB starts compressed zram swap. OpenVINO setup requires persistent USB storage or an installed system; RAM-only live storage cannot hold its runtime and models.
Optional Firefox: ooonana get firefox, then nonroot ooonana-firefox setup. Launcher downloads browser/runtime only after explicit setup; setup/update refuse volatile live storage by default. Disposable nonroot Flatpak install/update, first-run and file-picker journeys passed. Installed-session acceptance remains separate; Firefox is not preinstalled in fresh WSL.
Ooonana OpenVINO Chat 0.2.1 has a browser GUI through an authenticated loopback bridge. Windows-only computer-control tools are unavailable on Linux.
Wi-Fi supports personal and enterprise profiles. NetworkManager, BlueZ, D-Bus, Intel Wi-Fi/Bluetooth firmware, Chromium, Python 3, sudo, su, and doas are included in full-i3.

Desktop and hardware (continued)

Disposable VM installation, password login and sudo passed as UID 1000. Physical Xorg, wireless, audio, sensors and inference remain separate checks. No audible testing is performed automatically.
OpenVINO Linux dependencies use exact wheel hashes and a pinned Ubuntu image/APT snapshot. Signing private key stays local; public CI signing is deferred.

Bootable PDF

docs/ooonana.pdf uses native RISC-V64 Linux 6.18.37 and BusyBox 1.37.0. Embedded VM boot, stable input, native/virtual Backspace, version and package sync passed. Actual Chromium PDF interaction remains a manual gate.
RV64 JavaScript emulator clock is scaled down 16x for CPU progress. Guest time is slower than real time. This terminal PDF is not the x86 i3 desktop.
Boot status updates after kernel warnings. A sequential RAM seed replaces per-file boot copying; the interactive shell and package backend use RAM. Keep the PDF tab visible. Slow viewers may need several minutes; wait for ooonana# before typing.
Latest shipped-code Node suite passed in 87 seconds, not a viewer guarantee. Metadata snapshot reuse reduced measured package list from 130 to 45 seconds and sync from 86 to 57; signatures/checksums remain verified. Emulator time CSR now follows CLINT with privilege checks.
docs/ooonana-lite.pdf is a legacy artifact, not the current optimized runtime.

Persistence maintenance

bunana --shutdown and --restart request orderly init shutdown, stopping writers before sync, swapoff and unmount/remount read-only. No forced fallback. Live storage warnings are silent; no files are deleted automatically.
ooonana-persistence backup requires explicit read-only ext4 mount, matching UUID and a new destination on another filesystem. Ownership, hardlinks and xattrs are verified. Active live overlay operations are refused.
ooonana-persistence migrate requires matching verified backup and explicit data-only confirmation. It atomically switches to a fresh base, keeps home data, and preserves the complete old overlay. Accounts, packages and system configuration must be recreated; see docs/persistent-usb-safety.md.

Build proof markers

OOONANA_CLI_OK
OOONANA_BOOT_OK
OOONANA_INSTALL_OK
OOONANA_FULL_I3_OK
More detail lives in README.md, docs/ooonana-ai.md, and docs/jarvis-agi-research.md.